

Bayesian Nonparametric Joint Modeling Framework

Multi-stage TAAF Data: $\mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_L\}$, per-unit $(z_{ij}, \tau_{ij}, y_{ij}, \delta_{ij})$

Dependent Dirichlet Process

$$G_I = \sum_{k=1}^{\infty} \pi_{Ik} \delta_{\theta_k^*}$$
$$\beta_{Ik} = \rho_I \beta_{I-1, k} + (1 - \rho_I) \tilde{\beta}_{Ik}$$

Gaussian Process Degradation

$$z_{ij}(t) = \eta_{ij} + \gamma_{ij}t + \xi_{ij}(t)$$
$$\kappa_{\xi}(t, t') = \omega_{\xi}^2 \exp(-|t - t'|^2 / 2\phi_{\xi}^2) + \nu_{\xi}^2$$

Joint Model (Shared RE)

$$w_{ij} \mid \gamma_{ij}, c_{ij} = k \sim \mathcal{N}(\mu_k^* + \psi \gamma_{ij}, \sigma_k^{2*})$$

Hybrid MCMC: Slice Sampling + Blocked Gibbs + MH + HMC/NUTS

$\hat{f}_I(w)$

$\hat{R}_I(t)$

$\{\rho_I\}, \psi$

HPD intervals